AIM

To implement subnetting and Variable Length Subnetting Mask (VLSM) in a network using Cisco Packet Tracer and verify communication across subnets.

OBJECTIVES

1. Understanding IPv4 addressing, CIDR notation, subnetting, and VLSM in an IPv4 n/w.
2. Assign subnet addresses to multiple LANs based on host requirements.
   (a) Using fixed-length subnets (b) Using VLSM.
3. Configure routers and PCs in Cisco Packet Tracer.
4. Verify intra-subnet and inter-subnet communication using ping commands.

THEORY

- Classless IPv4 addressing: It allows networks to be divided flexibly using a custom prefix length to define the size of a network.
- CIDR notation: CIDR (Classless Inter-Domain Routing) represents an IP address with a suffix to indicate the no. of bits used for the network portion.
- Subnetting: A technique to divide a larger network into smaller subnets.
- VLSM (Variable Length Subnet Masking): An advanced form of subnetting, VLSM assigns different subnet sizes based on specific needs.

OBSERVATIONS

Part - 1 : Fixed Length Subnetting

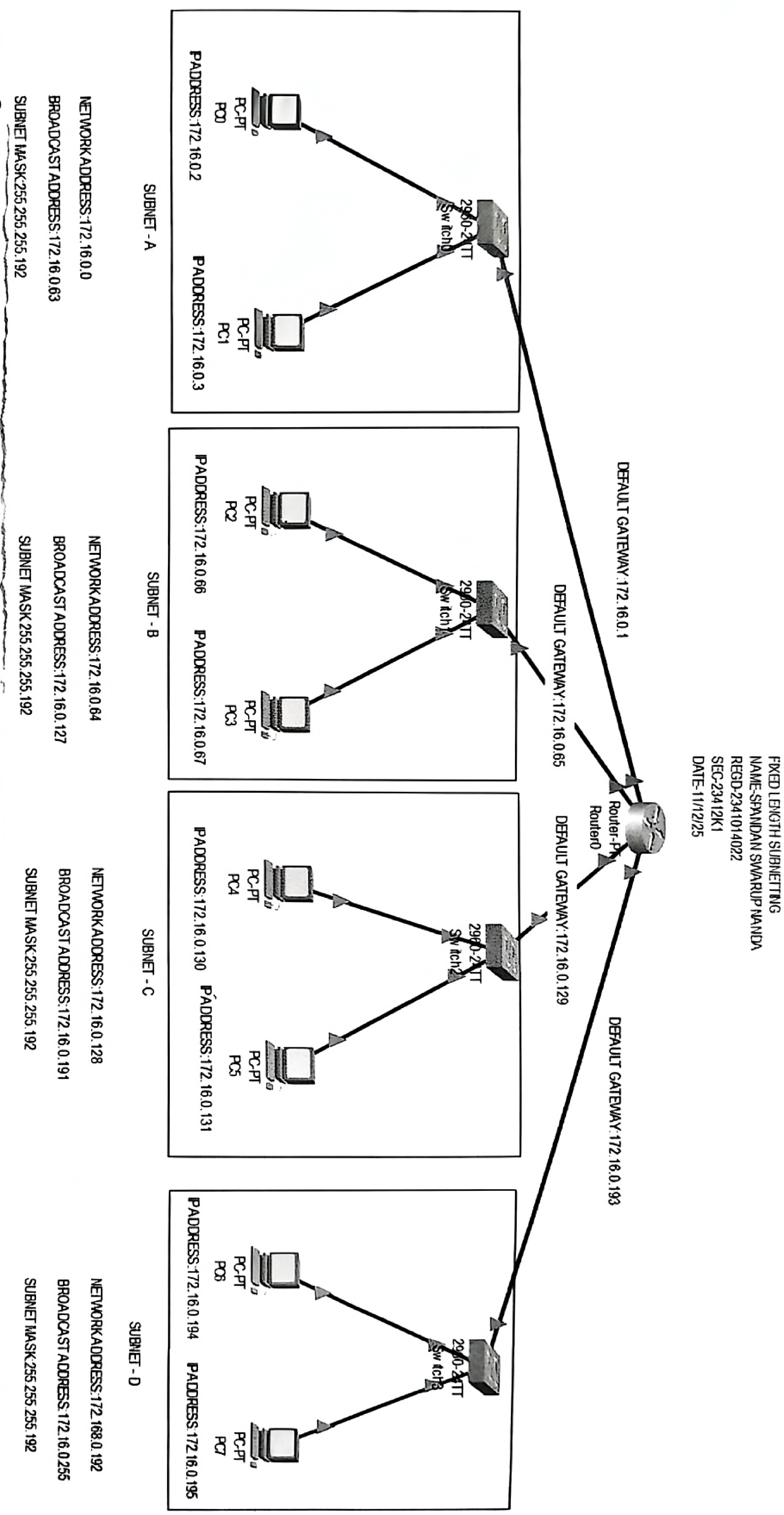

Fig. Packet Tracer simulation demonstrating the division of 172.16.0.0 into four subnets using FLSM. Accurate implementation of /26 fixed-length subnetting on a Class B networks.

- Inter-ping

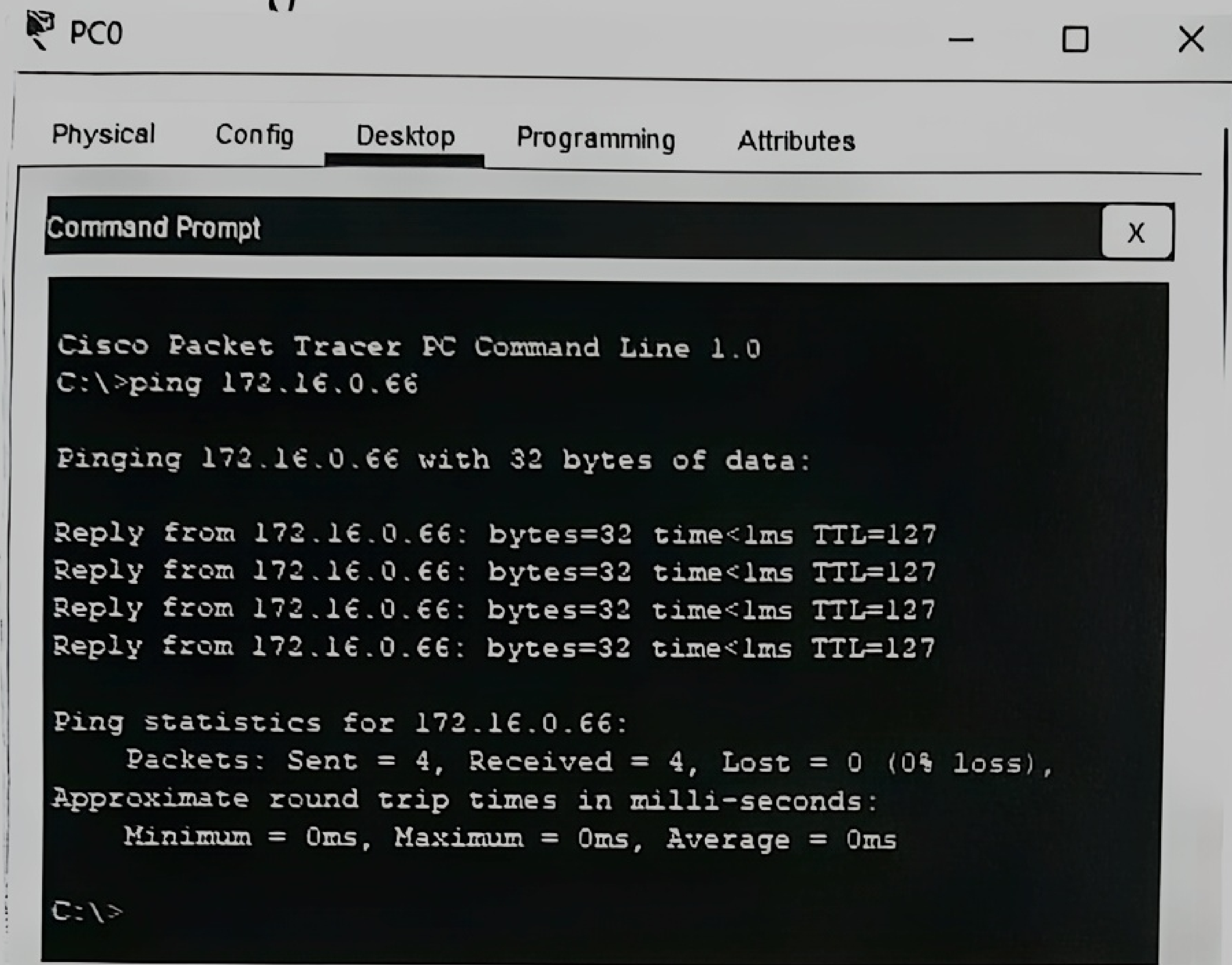

Fig. Command prompt showing a successful ping from PC0 to PC2 (172.16.0.66)
Confirms functional inter-subnet routing between Subnet A and Subnet B.

- Intra Ping

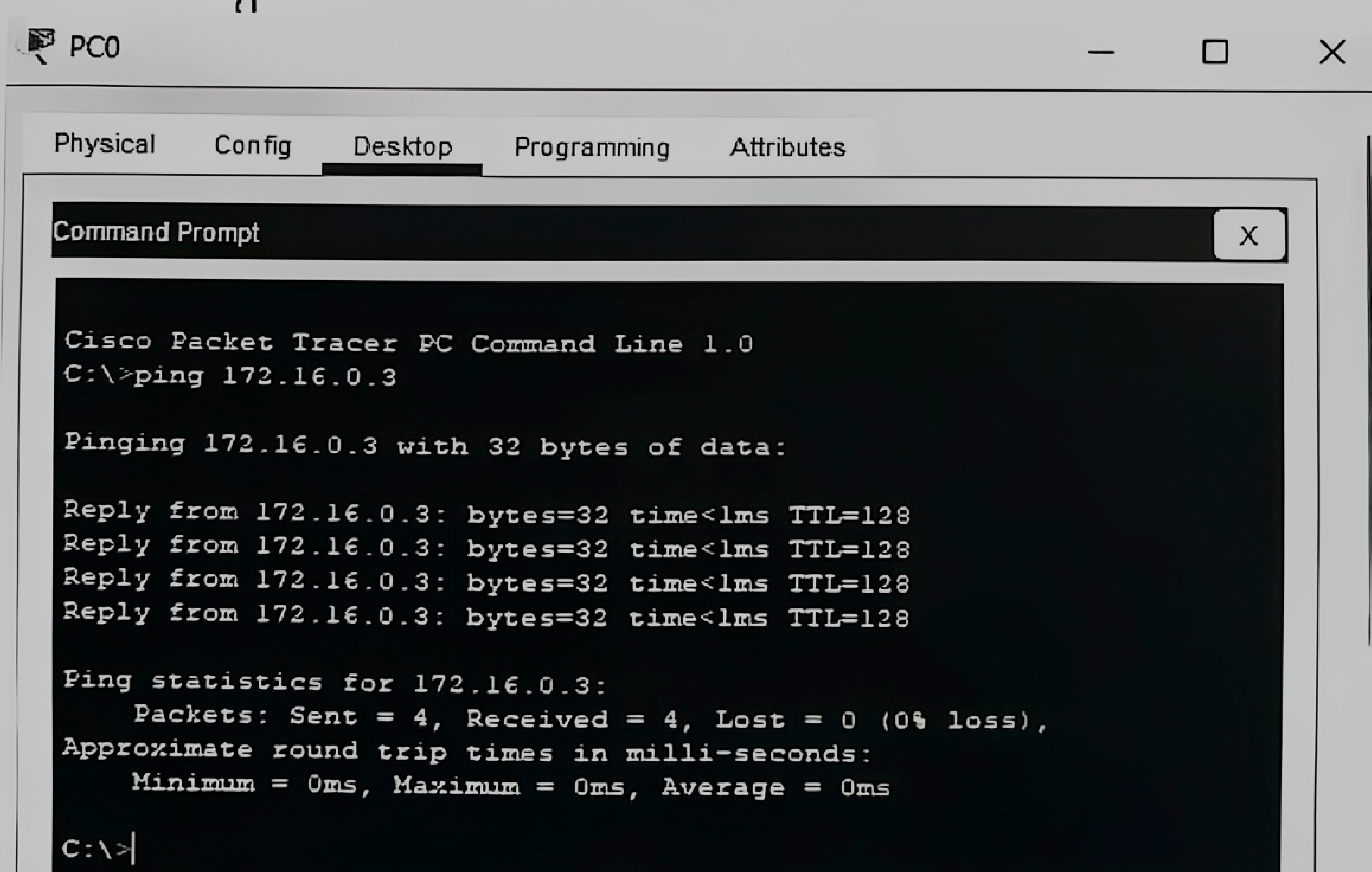

Fig. Command prompt showing a successful ping from PC0 to PC1 (172.16.0.3) within the same subnet
Verifies local Layer 2 switching connectivity with Subnet A.

Part 2: VLSM

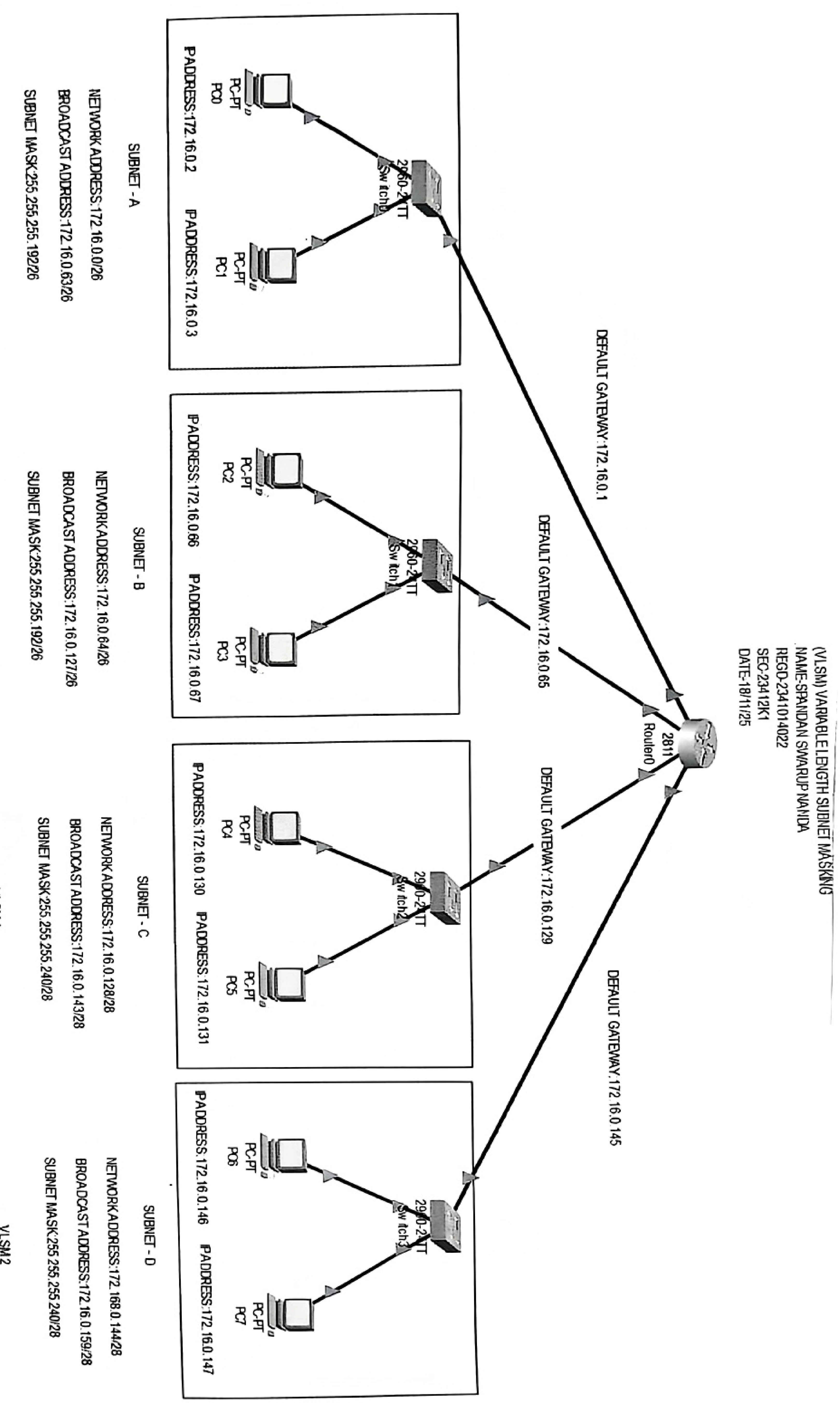

Fig. Packet Tracer VLSM Topology
A clear implementation of Variable length subnet Masking demonstrating efficient IP allocation by mixing /26 and /28 subnets.

• Inter-ping

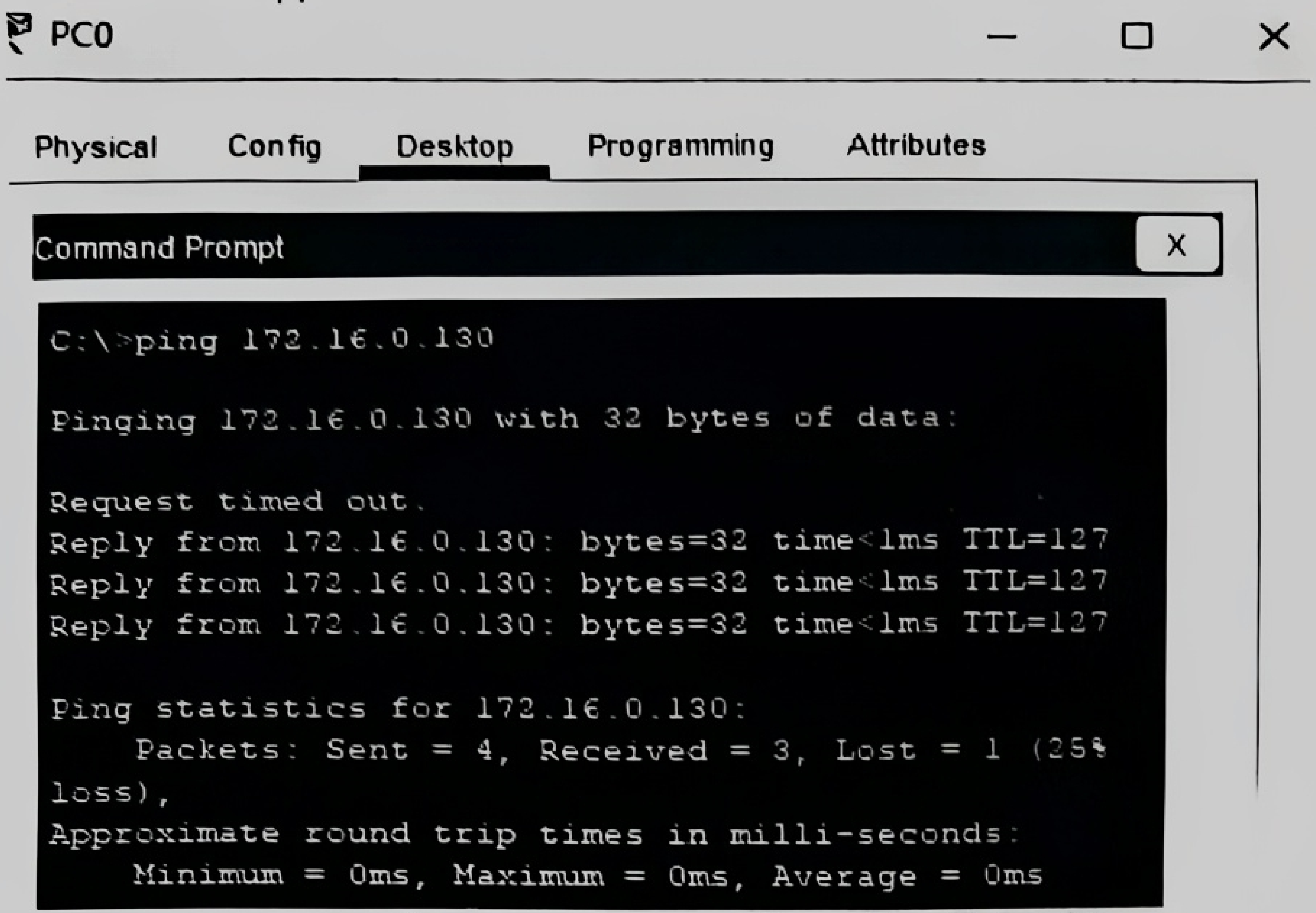

Fig. Successful inter subnet ping verification
The replies confirm active routing to Subnet C, with the initial timeout caused by normal ARP resolution delay.

• Intra-ping

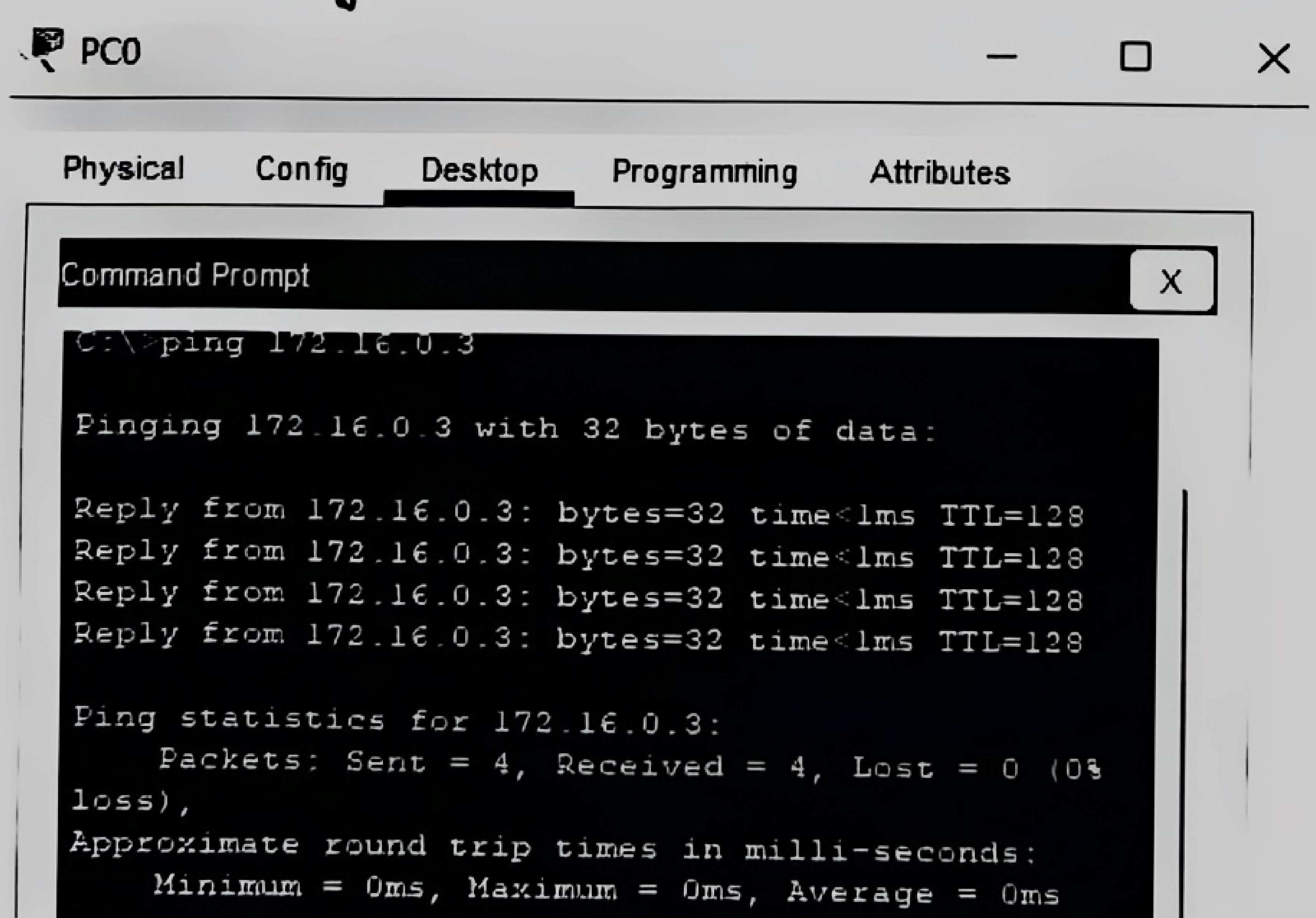

Fig. Successful Intra-Subnet Ping Verification.
All packets were received successfully with TTL = 128, confirming direct local connectivity (Layer 2 switching) without router intervention.

• ANALYSIS

Fixed Length Subnetting (FLSM)

- N/w 172.16.0.0/24 is divided into four equal subnets (/26)
- Each subnet has:
  - Same subnet mask: 255.255.255.192
  - Equal no. of usable hosts.
- Simplifies design but may waste IP addresses.

VLSM

- Subnets are created based on actual host requirements.
- Larger subnets (/26) for bigger LANs and smaller subnets (/28) for smaller LANs.
- Provides better IP address utilization compared to FLSM.

Intra-Subnets Ping Results

- Ping b/w the hosts within the same subnet is successful.
- eg: 172.16.0.2 ↔ 172.16.0.3
- Shows correct IP configuration and switching
- TTL values remain high, indicating local delivery.

Inter Subnet Ping Results

- Ping b/w hosts in different subnets is mostly successful
- eg: 172.16.0.2 ↔ 172.16.0.66
- Router correctly forwards packets using default gateways.
- Confirms Layer 3 routing ~~persona~~ functionality.

Gateway and Routing Verification

- Each subnet uses a default gateway on the router.
- Router interfaces correctly mapped to subnet n/ws.
- Ensures end-to-end communation across LANs.

CONCLUSION

In this experiment, IPv4 subnetting concepts using both fixed-length subnetting and VLSM were successfully implemented in Cisco Packet Tracer. Subnet addresses were correctly assigned to multiple LANs based on host requirements, and routers and PCs were properly configured. Successful intra-subnet and inter-subnet communication verified through ping commands confirms correct addressing, gateway configuration and routing. The experiment demonstrates that VLSM provides more efficient IP address utilization compared to fixed length subnetting.

EXERCISES

1. Express the following classful IP addresses in CIDR notation:
a) 195.30.10.20 b) 100.50.20.10 c) 130.10.15.25

2. Given the IP address of a device as 192.169.20.126/25. Find the subnet mask and network ID in dotted decimal notation.

3. A network with ID 200.10.20.0 is divided into 4 subnets. Find the number of hosts per subnet. Also, for all the subnets, find
a) Subnet Address b) First Host ID c) Last Host ID d) Broadcast Address

4. Design a network using VLSM for the following requirements with the given network 200.100.50.0/24. Assign IP addresses accordingly: (a) Network A: 50 hosts, (b) Network B: 25 hosts, (c) Network C: 15 hosts, (d) Network D: 10 hosts.

(1)(a) 192.30.10.20
- Class C → Default mask /24
- CIDR: 195.30.10.20/24

(b) 100.50.20.10
- Class A → Default mask /8
- CIDR: 100.50.20.10/8

(c) 130.10.15.25
- Class B → Default mask /16
- CIDR: 130.10.15.25/16

(2) Given IP: 192.169.20.126/25
- Subnet Mask: 255.255.255.128
- Network ID: 192.169.20.0

(3) Original Class C → /24
- 4 subnets → /26
- Hosts per subnet: $2^6 - 2 = 62$

Subnet	Subnet Address	First Host	Last Host	Broadcast
1	200.10.20.0	200.10.20.1	200.10.20.62	200.10.20.63
2	200.10.20.64	200.10.20.65	200.10.20.126	200.10.20.127
3	200.10.20.128	200.10.20.129	200.10.20.190	200.10.20.191
4	200.10.20.192	200.10.20.193	200.10.20.254	200.10.20.255

(4) VLSM Design for 200.100.50.0/24

Network	Hosts	Subnet	Network Address	Usable Host Range	Broadcast
A	50	/26	200.100.50.0	200.100.50.1 - 62	200.100.50.63
B	25	/27	200.100.50.64	200.100.50.65 - 94	200.100.50.95
C	15	/28	200.100.50.96	200.100.50.97 - 110	200.100.50.111
D	10	/28	200.100.50.112	200.100.50.113 - 126	200.100.50.127

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